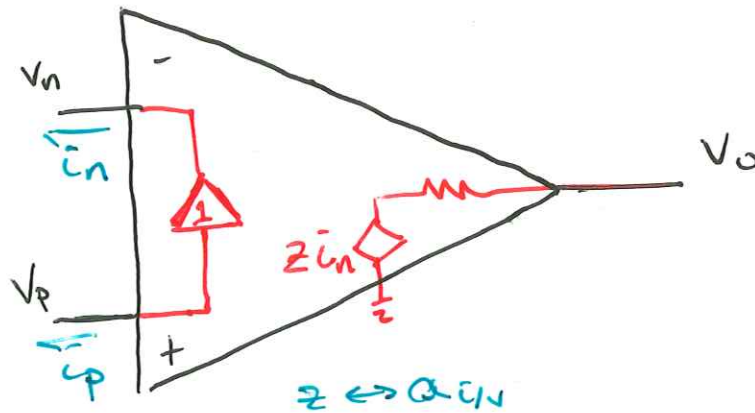


Current feedback Amplifiers (CFAs)



Open loop gain:

$$V_o = z \bar{i}_n \quad (\text{assuming } i_p \rightarrow 0)$$

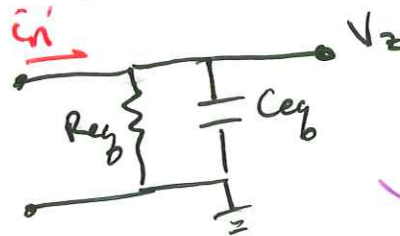
Amplification not directly dependent on V_p or V_n !

$$\text{Gain: } \frac{V_o}{\bar{i}_n} = z(jf)$$

$$\begin{aligned} i_p &= 0 \\ i_n &\neq 0 \\ V_p &= V_n \end{aligned}$$

what is $z(jf)$?

gain stage



$\bar{i}_n \rightarrow$ mirrored instance of i_n

$$V_z = z(jf) \bar{i}_n$$

$$V_o = z(jf) \bar{i}_n$$

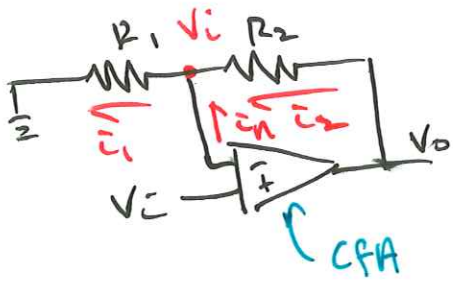
$$\therefore V_z = V_o$$



$$z(jf) = R_{cg} \parallel C_{cg} =$$

$$\frac{R_{cg}}{1 + j\omega R_{cg} C_{cg}} = \frac{R_{cg}}{1 + jf/f_b}$$

$$f_b = 1/2\pi R_{cg} C_{cg}$$



$$\bar{i}_1 = V_i / R_1 \quad \bar{i}_2 = \frac{V_o - V_i}{R_2}$$

$$\bar{i}_n + \bar{i}_2 = \bar{i}_1 \rightarrow \bar{i}_n = \bar{i}_1 - \bar{i}_2$$

$$\bar{i}_n = V_i \left[\frac{1}{R_1} + \frac{1}{R_2} \right] - \frac{V_o}{R_2}$$

$$V_o = z(jf) \left[V_i \left(\frac{1}{R_1} + \frac{1}{R_2} \right) - \frac{V_o}{R_2} \right]$$

$$\frac{V_o}{V_i} = \frac{\frac{1}{R_1} + \frac{1}{R_2}}{\frac{1}{R_2} + \frac{1}{z(jf)}} = A(jf)$$

want: $A(jf) = \frac{A_o}{1 + 1/T(jf)}$

$f \rightarrow 0$
if T large, $A_o \approx A_{ideal}$

$$\frac{V_o}{V_i} = \frac{1 + \frac{R_2}{R_1}}{1 + R_2/z(jf)}$$

$$T(jf) = z(jf)/R_2$$

similar to:

$$a(jf) \beta$$

$$\beta = 1/R_2$$

$$A_{ideal} = 1 + \frac{R_2}{R_1}$$

force T large at $f \rightarrow 0$:

$$z(jf) \Big|_{f \rightarrow 0} \rightarrow R_{eq} \quad T(jf) \Big|_{f \rightarrow 0} = R_{eq}/R_2$$

$\rightarrow R_{eq}$ must be large, $R_2 \ll R_{eq}$

with $R_2 \ll R_{eq}$

$$\frac{V_o}{V_i} = \frac{1 + R_2/R_1}{1 + R_2/z(jf)}$$

$$z(jf) = R_{eq} / (1 + jf/f_b) = z_o / (1 + jf/f_b)$$

$$= \frac{1 + R_2/R_1}{1 + \cancel{R_2/z_o} + j \frac{R_2}{z_o} \frac{f}{f_b}}$$

$\hookrightarrow R_2 \ll z_o \rightarrow 0$

$$A_o = \frac{1 + R_2/R_1}{1 + R_2/z_o} \quad \left\{ \begin{array}{l} z_o = R_{eq} \end{array} \right.$$

$$= \frac{1 + R_2/R_1}{1 + j \frac{R_2}{z_o} \frac{f}{f_b}} = \frac{1 + R_2/R_1}{1 + jf/f_b}$$

$$f_b = f_{b0} z_o / R_2$$

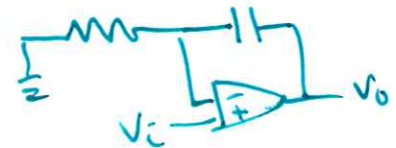
$\hookrightarrow$ bandwidth only dependent on R_2 !

$\rightarrow$ first set bandwidth of circuit by selecting R_2
(mindful of datasheet specs.)

$\rightarrow$ set gain by selecting R_1

considering $R_2 \ll R_{eq}$, $f \rightarrow 0$
recall $C \rightarrow \infty$ at $f \rightarrow 0$

will cause instability!



CFA (non-inv.) with values:

$$R_{eq} = 1M\Omega, C_{eq} = 1pF$$

$$Z_0 = 1 \text{ M}\Omega = 1 \text{ mV/A}$$

open loop - $V_o = 5V$ (just arbitrary consideration)
 $f \rightarrow 0$

$$V_0 = 5V = z_0 \bar{i}_n \rightarrow \bar{i}_n = 5 \mu A$$

if $R_1 \dot{\leq} R_2$ are $\leq z_0$, Reg

$$i_1, i_2 \Rightarrow i_n$$

→ $\bar{c}_n$ mostly negligible compared to $\bar{c}_1, \bar{c}_2$

→ $i_n \approx 0$
 $i_p = 0$
 $v_n = v_p$ } same as in VFA but for different reasons!

Pros: - Can achieve high bandwidth + high gain together

- Bandwidth easily set by R_z .

Cons:

Cons:

- Voltage / current offsets worse.

- lower open-loop gains

- May be noisier due to higher effective bandwidths